

Customer Churn Prediction Using Machine Learning

Data Analytics Internship
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1. Executive Summary

This project develops a machine-learning workflow for predicting customer churn using the IBM Telco Customer Churn dataset. The work covers data cleaning, exploratory data analysis, categorical encoding, train-test splitting, Logistic Regression, Random Forest, and XGBoost classification. The models are evaluated using accuracy, recall, and ROC-AUC. Customer-level churn probabilities and XGBoost feature importance are also examined.

2. Problem Statement

Customer churn is a binary classification problem in which the objective is to identify customers who may stop using a service. This project uses demographic, service, contract, tenure, and billing information to model the Churn outcome.

3. Objectives

- Understand observed churn patterns in the dataset.
- Perform data cleaning and exploratory data analysis.
- Prepare categorical and numerical variables for machine learning.
- Build Logistic Regression, Random Forest, and XGBoost models.
- Evaluate models using accuracy, recall, and ROC-AUC.
- Generate customer-level churn probabilities and interpret feature importance.

4. Dataset Description

Item	Details
Dataset	IBM Telco Customer Churn dataset
Records	7,043 customers
Original columns	21
Model features	19 after removing customerID
Target	Churn (0 = No, 1 = Yes)
Observed churn rate	26.54%

5. Data Cleaning & Preprocessing

- Loaded the CSV dataset using Pandas.
- Checked data types, missing values, and duplicate records.
- Converted TotalCharges from text to numeric.
- Found 11 blank TotalCharges values and replaced them with 0.
- Removed customerID because it is an identifier.
- Converted Churn from Yes/No to 1/0.

- Used one-hot encoding for categorical variables.
- Split data into 80% training and 20% testing using stratification.

6. Exploratory Data Analysis

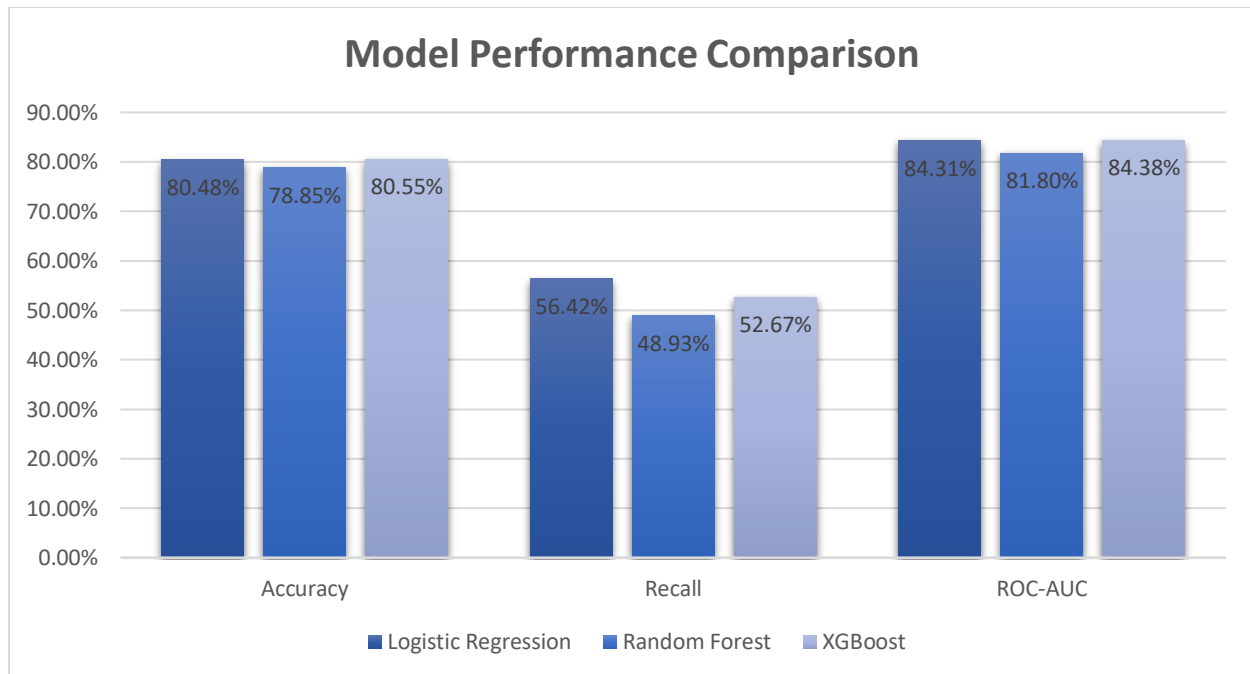
Analysis	Observed result
Overall churn	26.54% of customers churned.
Contract	Month-to-month customers showed higher observed churn.
Internet service	Fiber optic: 41.89%; DSL: 18.96%; no internet service: 7.40%.
Payment method	Electronic check: 45.29%; credit card (automatic): 15.24%.
Technical support	No Tech Support: 41.64%; Yes: 15.17%.
Monthly charges	Average: 74.44 for churned vs 61.27 for retained.
Tenure	Shorter-tenure customers showed more observed churn than long-tenure customers.

7. Machine Learning Methodology

Three classification algorithms were trained using the same preprocessing pipeline and held-out test set: Logistic Regression, Random Forest, and XGBoost. Categorical variables were one-hot encoded, while numerical variables were passed through.

8. Model Evaluation

Model	Accuracy	Recall	ROC-AUC
Logistic Regression	80.48%	56.42%	84.31%
Random Forest	78.85%	48.93%	81.80%
XGBoost	80.55%	52.67%	84.38%



9. Confusion Matrix & ROC Analysis

For Logistic Regression, the test-set confusion matrix was $\begin{bmatrix} 923 & 112 \\ 163 & 211 \end{bmatrix}$, corresponding to 923 true negatives, 112 false positives, 163 false negatives, and 211 true positives. ROC-AUC values were 0.843, 0.818, and 0.844 for Logistic Regression, Random Forest, and XGBoost respectively.

10. Features Importance

- Contract: Month-to-month was the most prominent feature in the XGBoost feature-importance output.
- InternetService: Fiber optic was another prominent feature.
- TechSupport: No and OnlineSecurity: No were among the higher-importance encoded features.
- Tenure and payment/service-related variables also contributed to model predictions.

Feature importance describes model usage of variables and does not establish causation.

11. Customer-Level Prediction

The XGBoost model generated churn probabilities for customers in the held-out test set. Several displayed test customers received predicted churn probabilities above 90%, demonstrating customer-level risk scoring.

12. Key Business Insights

- Observed churn rate was 26.54%.
- Month-to-month contracts showed higher observed churn than longer contracts.
- Fiber optic customers had a 41.89% observed churn rate.
- Electronic check customers had a 45.29% observed churn rate.
- Customers without technical support had a 41.64% observed churn rate.
- Churned customers had higher average monthly charges than retained customers.
- Contract and service-related variables were prominent in the XGBoost feature-importance output.

13. Limitations

- The dataset is a public historical dataset and may not represent every telecom business or current customer population.
- The analysis identifies associations and predictive patterns, not causal effects.
- Reported metrics depend on the selected train-test split and model settings.
- Further validation, threshold tuning, calibration, and monitoring would be appropriate before operational deployment.

14. Future Scope

- Tune model hyperparameters using cross-validation.
- Evaluate precision, F1-score, calibration, and threshold-specific performance.
- Experiment with class weighting or threshold tuning when recall is a priority.
- Create a Power BI or Streamlit dashboard for customer-risk monitoring.
- Validate the model on newer or organization-specific data.

15. Conclusion

The project demonstrates an end-to-end customer churn prediction workflow. On the held-out test set, all three models achieved ROC-AUC values above 0.80. Logistic Regression achieved 80.48% accuracy, 56.42% recall, and 0.8431 ROC-AUC. XGBoost achieved 80.55% accuracy, 52.67% recall, and 0.8438 ROC-AUC. Random Forest achieved 78.85% accuracy, 48.93% recall, and 0.8180 ROC-AUC. These results describe performance on this dataset and test split and should not be interpreted as causal conclusions.